Decode Labs Project 1 Report

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Introduction

This report documents the findings and methodology of Project 1 undertaken as part of the Decode Labs Data Science Internship. The project, titled Advanced EDA and Feature Engineering, forms the foundational milestone of the internship programme, requiring interns to demonstrate the ability to transform raw, unstructured data into a mathematically clean dataset ready for machine learning algorithms.

The project follows the Input-Process-Output (IPO) architecture prescribed by Decode Labs, covering three phases, securing input fidelity through missing value imputation and outlier neutralisation, processing through vectorised computation and encoding and outputting validated feature stores with runtime schema contracts.

Dataset Overview

The dataset used for this project is an e-commerce orders dataset consisting of 1,200 rows and 14 columns. It captures transactional records including product details, customer information, pricing, order status, payment methods and referral sources.

Missing Data Analysis and Imputation

The first step in securing input fidelity was identifying missing values across all 14 columns. The analysis revealed that only one column contained missing data which was Coupon Code with a missing count of 309.

According to the Missing Data Decision Matrix from the project framework, a missingness rate above 20% for numeric columns would trigger K-Nearest Neighbours (KNN) imputation. However, CouponCode is a categorical column, it is missing not because the data was lost, but because those customers did not use a coupon. Filling with 'No Coupon' is therefore the statistically correct and semantically meaningful decision, preserving the natural distribution of the data. After imputation, zero missing values remained across all 1,200 rows and 14 columns.

Outlier Detection and Neutralisation

The Interquartile Range (IQR) method was applied to all four numeric columns. The IQR provides a robust, non-parametric boundary to isolate statistical anomalies without assuming a normal distribution.

Winsorisation (numpy.clip) was used to neutralise outliers by capping values at the boundary rather than deleting rows. This preserves all 1,200 rows and maintains sequential data integrity.

Only TotalPrice contained genuine outliers, 8 orders with values exceeding R3,330.41. These were capped at the upper boundary. No rows were removed, ensuring the full dataset remained available for downstream modelling.

Feature Engineering

Three new predictive features were engineered from existing columns to add signal value for machine learning models. Feature engineering is the process of using domain knowledge to create new variables that better represent the underlying patterns in the data.

RevenuePerItem was subsequently found to be perfectly collinear with UnitPrice (correlation = 1.000) and was removed during the collinearity check in Step 7. IsHighValueOrder resulted in a perfectly balanced split: 600 high-value orders and 600 standard orders. HasCoupon revealed that 891 of 1,200 orders (74.25%) used a discount coupon.

Categorical Encoding

Machine learning estimators operate on real-numbered coordinate spaces and cannot interpret text labels. Four categorical columns were converted into binary numeric columns using One-Hot Encoding (OHE) with drop_first=True to eliminate the dummy variable trap and prevent multicollinearity.

Column Encoded	Unique Categories	New Columns Created
Product	7 (Chair, Desk, Laptop, Monitor, Phone, Printer, Tablet)	6
PaymentMethod	5 (Cash, Credit Card, Debit Card, Gift Card, Online)	4
OrderStatus	5 (Cancelled, Delivered, Pending, Returned, Shipped)	4
ReferralSource	5 (Email, Facebook, Google, Instagram, Referral)	4

The dataset expanded from 17 columns (after feature engineering) to 31 columns after encoding. Each category was mapped to an orthogonal coordinate axis, ensuring equal geometric distance between all categories in the feature space.

Collinearity Analysis

Multicollinearity occurs when two or more predictor variables are highly correlated, causing the feature matrix to become singular and non-invertible. This destabilises coefficient estimates and destroys model generalisation. The collinearity eradication algorithm from the project framework was applied in four steps:

- Step 1: Build the absolute Pearson correlation matrix for all numeric features
- Step 2: Isolate the upper triangle to avoid duplicate pairs
- Step 3: Identify all pairs with correlation exceeding 0.80
- Step 4: Drop the feature with the weaker correlation to the target

RevenuePerItem equals TotalPrice divided by Quantity. Since UnitPrice already represents price per unit directly, RevenuePerItem carried no additional information and was correctly removed. No other collinear pairs were found across the remaining features.

Schema Validation with Pandera

Pandera was used to enforce runtime structural contracts on the cleaned dataset, treating the data pipeline as a critical software interface. Lazy validation (lazy=True) was applied to collect all failures into a single diagnostic report rather than crashing on the first error.

Column	Contract Enforced	Validation Result
Quantity	Integer, value > 0	PASSED
UnitPrice	Float, value > 0	PASSED
TotalPrice	Float, value > 0	PASSED
IsHighValueOrder	Integer, value in {0, 1}	PASSED
HasCoupon	Integer, value in {0, 1}	PASSED
CouponCode	String (not null)	PASSED

All six contracts passed validation. The dataset meets the required structural integrity standards for downstream machine learning pipelines.

Key Findings and Business Insights

Data Quality

- The dataset was 97.7% complete, only CouponCode had missing values (25.75%), resolved through categorical fill.
- TotalPrice was the only column with genuine outliers (8 extreme values), all neutralised via Winsorisation.

- No rows were deleted throughout the entire pipeline, preserving all 1,200 records.

Product & Sales Patterns

- Printer was the most ordered product (181 orders), followed by Tablet (179) and Chair (178).
- Phone was the least ordered product (156 orders), a potential area for targeted marketing.
- The average order value was R1,053.64 with a median of R823.62 and a maximum of R3,330.41.

Customer Behaviour

- 74.25% of all orders (891 out of 1,200) used a discount coupon, indicating strong price sensitivity.
- Orders split exactly 50/50 between high-value (above R823.62) and standard value, a well-balanced target variable for classification models.
- Instagram was the top referral source (259 orders), suggesting strong social media acquisition.

Payment Methods

- Online payment was the most popular method (258 orders), closely followed by Cash (246) and Credit Card (234).
- All five payment methods had fairly even distribution, suggesting no dominant payment preference.

Order Status

- Order statuses were almost evenly distributed across Cancelled (250), Returned (247), Pending (237), Shipped (235), and Delivered (231).
- The high combined rate of Cancelled and Returned orders (497 out of 1,200, 41.4%) is a significant business concern worth investigating further.

Conclusion

Project 1 successfully demonstrated the full Advanced EDA and Feature Engineering pipeline on a 1,200-row e-commerce dataset. Following the Input-Process-Output architecture prescribed by Decode Labs, the raw dataset was systematically cleaned, validated and transformed into a mathematically sound feature store ready for machine learning.

The pipeline applied the Missing Data Decision Matrix to handle CouponCode missingness, used IQR-based Winsorisation to neutralise outliers in TotalPrice without losing any rows, engineered three new predictive features, applied One-Hot Encoding across four categorical columns, detected and removed a perfectly collinear feature pair, and validated all structural contracts using Pandera, all passing.

The final clean dataset contains 1,200 rows and 16 columns, with two engineered binary features (IsHighValueOrder and HasCoupon) that are directly usable as predictors or target variables in downstream supervised learning models. The skills demonstrated in this project, statistical imputation, outlier neutralisation, feature engineering, and data contract validation, form the foundation of professional, production-ready data science work